FRC Sensor CAN B-2 · Factory instructions

Scope: Rev B-2. Revision overview · All revisions.

This package is ready to submit for a **prototype fabrication and turnkey assembly quotation**. Factory stock matching, panel/fixture review and placement-orientation approval remain necessary before manufacture. It is not a completed JLCPCB checkout or a qualified production release.

Files to submit

1. `order/RevB2_Gerbers.zip`: single-board Gerbers and plated/nonplated Excellon drills, in millimeters. Do not scale or mirror.

2. `bom/BOM_turnkey.csv`: complete exact-manufacturer/MPN sourcing BOM, including all 144 fitted parts and ten loose shunts per board.

3. `output/assembly/CPL_JLCPCB.csv`: all 144 fitted placements, top side. `bom/BOM_JLCPCB.csv` supplies the corresponding 50 fitted BOM groups. Blank LCSC fields require exact-MPN global sourcing or consignment; they do not mean DNP. The shunts are separately listed in `bom/ACCESSORIES.csv` and must be bagged.

4. `output/assembly/top_assembly.pdf`, `bottom_markings.pdf`, `schematic.pdf`, `pad_coordinates.csv` and the interactive BOM for pin-one and polarity review.

5. `output/mechanical/FRC_Sensor_CAN.step`, the 1:1 mechanical PDF, and these instructions. The STEP models are conservative placement aids, not certified connector tooling models.

PCB and panel requirements

Setting	Requirement
Finished board	60 × 68 mm, 2 layers, 1.6 mm FR-4
Copper	Nominal finished 35 µm / 1 oz per side; use the supplied electrical clearances
Finish	ENIG, both masks, green mask, white silkscreen
Minimum features	0.18 mm tracks; 0.20 mm clearance; ordinary vias 0.60/0.30 mm; preserve every supplied drill
Assembly	128 SMT and 16 through-hole components, all on top; no bottom components
Stencil	0.10 mm starting specification, supplied top paste; assembler to review aperture release and U3 exposed-pad paste
Pogo connector J8	Ten bare ENIG contacts: NO paste, solder, connector or conformal coating; three alignment holes remain NPTH
Mounts	Four 3.2 mm NPTH; do not fill or plate
Solder	Lead-free process, engineer-reviewed profile compatible with every supplied component

For JLCPCB **Standard PCBA**, the single PCB is below the published 70 × 70 mm minimum processing envelope. Request factory panelization with removable tooling rails and fiducials. A 5 mm rail allowance around each side gives a nominal **70 × 78 mm processing outline** before panel spacing/tabs. The assembler must finalize tooling-hole/fiducial/tab positions. Do not drill new tooling holes in the functional 60 × 68 mm board: there is routed copper near the edges.

The six horizontal connector bodies project beyond the top and bottom board edges. Assemble SMT while supported by rails, then depanel and install these connectors in a supporting fixture, or propose an equivalent process with sufficient overhang clearance. The finished assembly reserves **60 × 76.1 mm**, excluding cables; terminal height is **8.5 mm**, leads **3.5 mm** before trimming. Breakaway tabs and fixtures must not foul the housings, actuator travel, solder joints or nearby copper. Panel design is part of the factory's engineering

service, not a supplied production panel.

The selected Samsung 100 nF capacitor's reference sheet recommends a maximum 250 °C reflow peak for six seconds. Use a controlled profile within component limits (JLCPCB Standard lists 240 ±5 °C). Do not automatically apply Economic PCBA's listed 255 ±5 °C profile to this BOM. Solder THT parts afterward. U3's exposed thermal pad must be soldered and inspected; do not omit it or its thermal vias. The assembler must review paste loss into these vias; no via-in-pad filling process is presumed in the cost estimate.

Placement coordinates and orientation

The factory CPL origin is the **lower-left finished-board corner viewed from above**: X increases right, Y increases up, and all coordinates are positive millimeters. Rotations are KiCad top-side counterclockwise degrees. SMT parts use the native body-center origin. Through-hole header and terminal centers have explicit offsets from their KiCad pin-one origins; those offsets are recorded in `coordinate_offsets.csv`.

`pad_coordinates.csv` provides every numbered pad in the same coordinate frame and is the final reference for matching factory library orientation. Confirm U1–U8, Q1, Y1, every diode/LED and C1 polarity. Do not assume an LCSC library's zero-degree rotation matches KiCad. The original `positions_all_top.csv` uses KiCad's upper-left origin with negative Y and is provided only as a native reference; **do not mix it with the factory CPL**.

Parts that require particular attention

• J1–J3: **Weidmüller 1824420000**, LSF-SMT 3.50/02/90 3.5SN BK TU. J4–J6: **1824430000**, LSF-SMT 3.50/03/90 3.5SN BK TU. These are horizontal wire entry, parallel to the PCB. Two legs per pole, 7 mm between rows, 1.1 mm drills. Do not substitute `/135`, `/180`, Phoenix or an unreviewed clone.

• U5: **STM32F042K6T6**, LQFP32. LCSC C69216 was out of stock at the observation date. Procure from DigiKey or another authorized source. Do not substitute STM32F030 or another MCU; pin compatibility alone is insufficient.

• U7: **LM339LVDR**, not an ordinary LM339. Its low-voltage input behavior is required.

• U4: **TPS2553DBVR-1**, including the `-1` latch-off suffix. U3: **TPS1H200AQDGNRQ1** with its exposed pad.

• Keep the specified inductors, fuse, TVS diodes, clamp diodes, timing/filter capacitors and crystal exact. Only the five part-family substitutions listed in `PART_CHANGES.md` are incorporated. Quote any further proposed change for engineering review before fitting it.

Every BOM reference is fitted unless explicitly labeled an accessory. J8, test pads and mounting holes are unpopulated PCB features. Deliver all ten configuration shunts loose in an antistatic bag. Ship power selectors, bias selectors, threshold selectors and CAN termination **open**. This leaves sensor-power connections open until the user configures the board.

Inspection, optional programming and delivery

Include bare-board electrical testing, SMT AOI, THT visual inspection, polarity inspection, and inspection of U3's hidden solder joint. Check that paired terminal legs connect to the same pole and adjacent poles remain isolated. Do not count bare-board testing/AOI as a functional CAN or sensor test.

The base assembly quote may ship an unprogrammed MCU. For a programmed board, separately quote flash/verify and basic power checks using `firmware/build/sensor_can.hex` or `.bin` (address **0x08000000** for BIN). File hashes are in the archive manifest. Read `firmware/README.md` and `WIRING.md`: J8 is a TC2050-IDC-NL **mechanical** pattern with a **custom electrical pinout**. Its pin 6 is target-voltage reference, not programmer power. Power the board through J1 with a current-limited supply; never inject probe 3.3 V into J8.

With all shunts removed, check the intermediate rail (~3.97 V), internal 5 V and logic 3.3 V. Flash, read back, verify and leave sensor rails disabled. Leave flash configuration erased unless an explicit per-unit ID provisioning list is supplied. A production fixture, full CAN/sensor/load tests and qualification are additional work; `QUALIFICATION.md` describes the required engineering measurements.

Pack assemblies individually in ESD bags, protect actuator overhang and pogo pads, include loose shunts and unused boards/components as agreed in the quotation. Return the exact fitted-MPN list and programming/inspection results with each batch.

Sources: JLCPCB assembly capabilities and reflow temperatures, JLCPCB pricing, Samsung capacitor reference sheet.

Reviewed component substitutions

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Rev B-2 changes procurement while preserving Rev B's circuit, 60 × 68 mm outline, horizontal connectors, copper routing, pin assignments and firmware. There are still 144 fitted components and ten loose shunts. No protection, filtering or decoupling part has been deleted.

Rev B part	Rev B-2 fitted part	Count	Review basis
Yageo RC0805FR-071KL	UNI-ROYAL 0805W8F1001T5E, C17513	7	1 kΩ, 0805, 1%, 125 mW, ±100 ppm/°C
Yageo RC0805FR-0710KL	UNI-ROYAL 0805W8F1002T5E, C17414	16	10 kΩ, same package/rating/tolerance/TCR
Yageo RC0805FR-074K7L	UNI-ROYAL 0805W8F4701T5E, C17673	7	4.7 kΩ, same package/rating/tolerance/TCR
Yageo RC0805FR-07100KL	UNI-ROYAL 0805W8F1003T5E, C149504	7	100 kΩ, same package/rating/tolerance/TCR
Yageo CC0805KRX7R9BB104	Samsung CL21B104KCFNNNE, C28233	12	100 nF, X7R, 10%, 0805; voltage rating increases from 50 V to 100 V

The resistor family is rated 150 V working voltage, with power derating above 70 °C. Original per-reference power and 1.75% total resistance budgets remain applicable to these 1%/100 ppm parts. The Samsung capacitor is 2.00 × 1.25 × 1.25 mm nominal and fits the existing 0805 lands; maximum height remains below the terminal envelope. Its effective capacitance, reflow profile and real circuit transients still need prototype validation. Changing rated voltage alone never establishes a converter's stability.

The original Yageo MPNs remain approved fallback parts if the assembler cannot source these alternatives. A fallback changes cost and must be recorded in the fitted BOM. No resistor value or dielectric may be changed as an informal substitution.

Eight additional MPNs stay electrically and physically identical but use cheaper observed LCSC procurement: TCAN3413DR, LMR51420XDDCR, TPS613222ADBVR, TPS2553DBVR-1, BAT46W-E3-18, CL31A226KAHNNNE, AO4485 and Littelfuse SMAJ30A. Remaining parts retain their exact DigiKey sourcing references. LCSC stock observations are not reservations or JLCPCB assembly-stock confirmations.

Keeping 0805 parts avoids a needless reroute and remains machine-assembly compatible. 0402 parts would not reduce the number of placements or distinct feeder setups. The premium Weidmüller connectors are retained as requested. Their DigiKey first-tier total is $11.25 per board; they are a substantial part of the cost.

There are 34 capacitors. Most are local bypass, regulator reservoir, crystal load, input filter or fault/reset timing components. C13, C24 and C42 are possible **bench evaluation** candidates for later reduction, not approved DNPs. See `PASSIVE_REVIEW.md` for every capacitor's purpose. Larger savings require functional circuit consolidation and hardware testing.

Primary specification sources: UNI-ROYAL chip resistor family, Samsung exact capacitor reference sheet. Supplier IDs, source URLs, observed quantities, MOQ and prices are recorded in `bom/catalog_china.json` and the turnkey BOM.